

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1710

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 2

Duration : 1 Hour
Total Marks:30

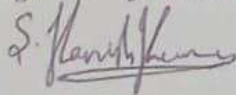
Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I Kanish Kumar S (192511415) (Name Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:






CO1
ASSESSMENT TOOL 2

1. A Courier Company must assign 3 delivery trucks (truck 1, truck 2, truck 3) to 3 delivery time-windows (9-11 AM, 11 AM-1 PM, 1-3 PM) such that,

CONSTRAINTS:

- (i) Truck 1 only cannot deliver in the 1-3 PM window.
- (ii) Truck 2 must deliver in before Truck 3.
- (iii) Only one truck per time-window.
- (iv) Truck 3 cannot deliver in the 9-11 AM window.
- (v) All trucks must be assigned exactly one-time window.

Apply Backtracking Search to find a Valid assignment, showing all intermediate steps and constraint checks. State the final valid schedule.

Ans)

PROBLEM:

Assign three delivery trucks (Truck 1, Truck 2, Truck 3) to three delivery time windows (9-11 AM, 11 AM-1 PM, 1-3 PM) while satisfying the given constraints.

VARIABLES:

T1 = Truck 1
T2 = Truck 2
T3 = Truck 3

DOMAINS:

9-11 AM
11 AM-1 PM
1-3 PM

CONSTRAINTS :

1. Truck 1 cannot deliver in 1-3 PM
2. Truck 2 must deliver before Truck 3
3. Only one truck can be assigned to each time window
4. Truck 3 cannot deliver 9-11 AM
5. Every truck must be assigned exactly one time window.

BACKTRACKING SEARCH STEPS :

STEP 1 :

Assign Truck 1 = 9-11 AM

Current Assignment :

Truck	Time
T1	9-11
T2	-
T3	-

Constraint check :

T1 is not in 1-3 PM ✓

STEP 2 :

Try Truck 2 = 9-11 AM

Constraint check :

Same time slot already assigned to T1 ✗

BACKTRACK

STEP 3 :

Assign Truck 2 = 11 AM-1 PM

Truck	Time
T1	9-11
T2	11-1
T3	-

Constraint check:

T2 is before T3 ✓

STEP 4:

Try Truck 3 = 9-11 AM

Constraint check:

Truck 3 cannot be in 9-11 AM X

BACKTRACK

STEP 5:

Try Truck 3 = 11 AM - 1 PM

Constraint Check:

Time slot already assigned to T2 X

BACKTRACK

STEP 6:

Assign Truck 3 = 1-3 PM

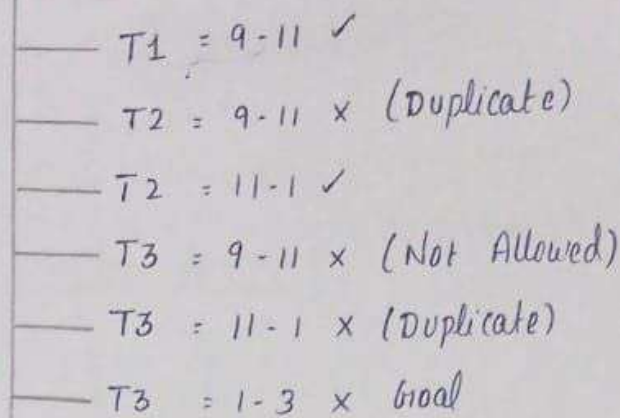
Truck	Time
T1	9-11
T2	11-1
T3	✓ 1-3

FINAL CONSTRAINT VERIFICATION:

- * T1 is not in 1-3 PM
- * T2 is before T3
- * One truck per time slot.
- * T3 is not in 9-11 AM.
- * Every Truck assigned exactly one slot.

BACKTRACKING SEARCH TREE:

Start



CONCLUSION:

Using the Backtracking Search algorithm, invalid assignments were rejected whenever a constraint was violated. After backtracking and checking all constraints, the valid assignment obtained is.

* Truck 1 → 9-11 AM

* Truck 2 → 11 AM-1 PM

* Truck 3 → 1-3 PM

2. A package-delivery robot navigates a 5x5 city-block grid from Start (S) to a delivery address (G). Certain streets are one-way, restricting movement direction in some cells.

Ans)

PROBLEM:

A Package delivery robot must travel on a 5x5 grid from Start (S) to Goal (G). The Manhattan Distance heuristic is used for estimating the distance to the goal.

GRID:

S	0	0	X	0
0	X	0	X	0
0	0	0	0	0
X	X	0	X	0
0	0	0	0	G

Legend;

S $\rightarrow$ Start

G $\rightarrow$ Goal

0 $\rightarrow$ open cell

X $\rightarrow$ Blocked cell

MANHATTAN DISTANCE HEURISTIC:

The Manhattan Distance is calculated as:

$$h(n) = |\text{Goal Row} - \text{Current Row}| + |\text{Goal Column} - \text{Current Column}|$$

$$\text{Goal} = (5, 5)$$

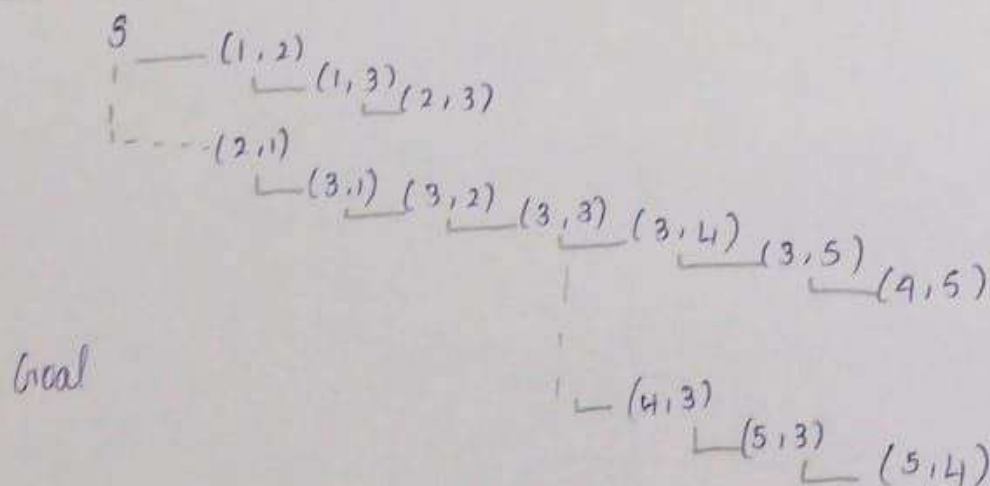
$$\text{Start} = (1, 1)$$

$$h(S) = |5 - 1| + |5 - 1| = 8$$

BFS ALGORITHM:

1. Insert the start node into the queue.
2. Remove the first node from the queue.
3. Expand all valid neighbouring nodes.
4. Mark Visited nodes.
5. Continue until the goal node is reached.

SEARCH TREE:



Node Expansion :

<u>STEP</u>	<u>EXPANDED NODE</u>	<u>FRONTIER QUEUE</u>
1	5	(1,2) (2,1)
2	(1,2)	(2,1) (1,3)
3	(2,1)	(1,3) (3,1)
4	(1,3)	(3,1) (2,3)
5	(3,1)	(2,3) (3,2)
6	(2,3)	(3,2) (3,3)
7	(3,2)	(3,3)
8	(3,3)	(3,4) (4,3)
9	(3,4)	(4,3) (3,5)
10	(4,3)	(3,5) (5,3)
11	(3,5)	(5,3) (4,5)
12	(5,3)	(4,5) (5,4)
13	(4,5)	(5,4) Goal
14	Goal	Search Ends

PATH COST :

Total Moves = 8
Cost per Move = 1
~~Total Cost = 8~~

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

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